

TermPlanMT: Engineering & Evaluation Log

Graph-Grounded FR→EN MT for Keytruda SmPC Section 4.8

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1 Overview

TermPlanMT translates Section 4.8 (*Undesirable Effects*) of the Keytruda (pembrolizumab) EMA SmPC from French to English. The section contains 127 aligned segments ($\approx 4,400$ words) and is selected for its high density of MedDRA-controlled terminology.

The pipeline connects to a Neo4j MedDRA graph (EN + FR), enabling terminology grounding, GraphRAG context injection, and decoding-level logit constraints. Five systems form a causal ablation ladder (Table 1).

ID	System	Graph	Ablation question
S1	NLLB-200 zero-shot	No	Lower bound
S2	Mistral 7B full-doc	No	Does long context help?
S3	Mistral + GraphRAG	Yes	Does retrieval alone help?
S4	GraphRAG + reranking	Yes	Does post-hoc enforcement add?
S5	GraphRAG + logit constraints	Yes	Is decoding-level control worth it?

Table 1: Experimental conditions.

2 S5 Architecture: Logit-Constrained Decoding

S5 is the only system that enforces terminology at **decoding time**. All other systems either ignore terminology (S1, S2), inject it as prompt context (S3), or rescore candidates post-hoc (S4). S5 modifies the model’s token probability distribution so that tokens belonging to MedDRA-preferred English strings are favoured at every generation step.

2.1 Pipeline Overview

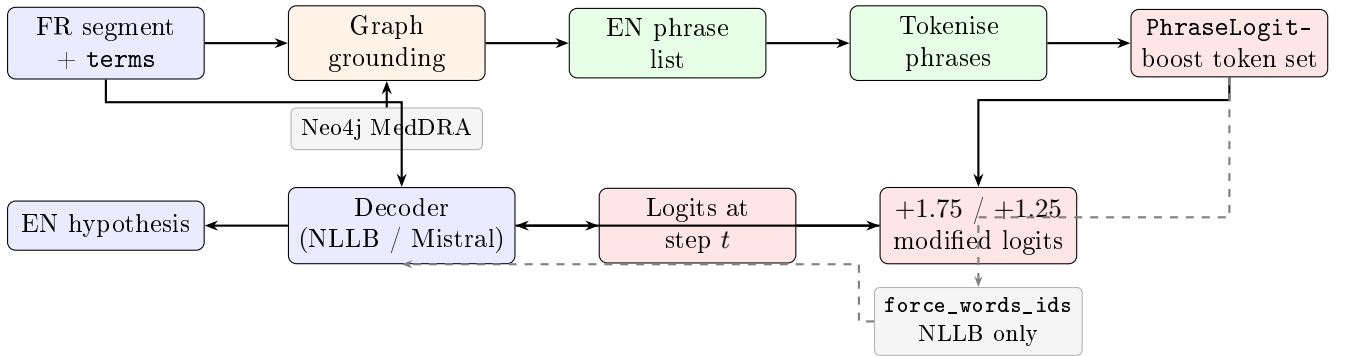


Figure 1: S5 data flow. NER spans are grounded via Neo4j to MedDRA concepts; their English names are tokenised and the resulting token ids receive a constant logit boost at every decoding step. NLLB additionally attempts `force_words_ids` (dashed) to hard-require phrases in beam search, falling back to soft boost on failure.

2.2 Step-by-Step

1. **NER spans.** Each segment carries a `terms` list from `segments_ner.jsonl` (CamemBERT-NER, score > 0.80). If `terms` is empty no boost fires and S5 behaves identically to S1/S3.
2. **Graph grounding.** Each French span is passed to `graph.ground()`: NFKC normalisation $\rightarrow$ exact match on `fr_label`; rapidfuzz fallback (≥ 90) on miss.
3. **English target selection.** If a planning lock exists for the span, its locked string is used; otherwise the concept’s canonical `name` (MedDRA preferred term at the matched level).

4. **Tokenisation.** Each English target is tokenised with the *active decoder’s own tokeniser* (NLLB sentencepiece or Mistral BPE). Every subword token id is collected into a set.
5. **PhraseLogitBoost at every step.** At each autoregressive step, +1.75 (NLLB) or +1.25 (Mistral) is added to the raw logit of every token id in the set before softmax. This makes constituent tokens more likely without forbidding others.
6. **NLLB force_words_ids (hard constraint, optional).** NLLB beam search additionally attempts to require each phrase verbatim. On failure (e.g. length conflict) it falls back to soft boost only.
7. **Mistral: S3 prompt + boost.** GraphRAG context is injected into the prompt exactly as in S3. The logit boost is the only addition.

2.3 Why Boost $\neq$ Hard Constraint

The boost is a *soft* intervention. A phrase will not appear if the model’s prior for an incorrect variant outweighs the boost. This explains why S5 and S3 produce nearly identical outputs on the pneumonitis cluster and immune-mediated modifier: “inflammatory pneumopathy” and “immunomodulatory” dominate the beam even at +1.75. The boost constant could be raised, but increasing it degrades fluency on segments where no gold term applies.

2.4 NLLB vs Mistral Backend

	NLLB	Mistral
Boost	+1.75	+1.25
Hard constraint	<code>force_words_ids</code> (fallback)	None
Prompt context	None (NMT encoder only)	GraphRAG + locks (as S3)
Speed	≈ 1.2 s/seg	≈ 6.1 s/seg
Output	<code>results/s5.jsonl</code>	<code>results/s5_mistral.jsonl</code>

Table 2: S5 backend comparison. NLLB uses a higher boost because the boost is its only MedDRA signal; Mistral has prompt context already so a lower boost avoids competing with it.

3 Infrastructure

3.1 Neo4j Graph

Built from MedDRA 29.0 EN + FR ASCII (non-commercial licence, MSSO/UPV-EHU).

Total concept nodes	120,910
Nodes with <code>fr_label</code>	93,185

Table 3: Graph statistics after full build.

Grounding (`src/graph.py`): NFKC normalisation $\rightarrow$ exact match on `fr_label`; rapidfuzz fallback (≥ 90) when exact match fails. Flag `GROUND_ENABLE_FUZZY=0` disables fuzzy for ablation.

3.2 NER Pipeline

NER spans are extracted by `scripts/prepare_data.py` and stored as `terms` fields in `data/section48/segments`.

Model: Jean-Baptiste/camembert-ner (trained classification head, no “MISSING classifier” warnings). Default threshold: `-min-ner-score 0.80`.

Coverage: 58 spans across 48/127 segments pass the threshold. Dataset CCR = 0.085.

Note: `almanach/camembert-bio-base` (original model) is a base LM with no NER head; scores never exceeded 0.58, yielding zero spans. `Dr-BERT/DrBERT-7GB` rejected for same reason plus pipeline crash.

3.3 Hardware

NVIDIA RTX 5060 Ti (8 GiB VRAM), Fedora Linux, Python 3.14. Mistral 7B quantised 4-bit (bitsandbytes NF4). Neo4j in Docker.

4 Metrics

BLEU / **chrF** Surface fluency (sacrebleu).

COMET Neural quality (model not yet downloaded; pending).

HTM *Hierarchy-aware Terminology Metric* (novel). For each (segment, gold term) pair where the French gold string appears in the source, checks whether any accepted EN rendering appears in the hypothesis. Hit: MedDRA-level score 0.0–1.0. Miss: 0.0. Macro-averaged over triggered pairs. **Penalises every missed rendering.**

TCR *Term Consistency Rate*. Gold entry must trigger (≥ 1 segment contains the FR string) and ≥ 1 hypothesis must contain an accepted EN rendering. If no hypothesis hits any rendering $\rightarrow 0.0$ (failure). If ≥ 1 hit $\rightarrow 1.0$ iff all canonical labels agree, else 0.0. Macro-averaged over applicable gold entries. **Punishes complete misses and inconsistent renderings; ignores per-segment recall.**

CCR *Concept Coverage Rate*. Fraction of NER-extracted FR spans grounded in Neo4j. Dataset-level property, not per-system.

Eight hand-curated gold entries in `gold_terms.json` drive HTM and TCR.

5 Results

Note: Results are reported under three NER regimes (Table 5). The primary results table uses BioMistral-7B NER (CCR = 0.354), which is the strongest NER condition. COMET is excluded from the paper table as the spread is 0.021 across all systems. The recommended main figure is `comparison_bubble_chrF_HTM_time.{pdf,png}`: x = chrF (fluency), y = HTM (terminology), bubble area = mean inference time.

5.1 Key Observations

Confirmed run: Full pipeline on BioMistral NER segments completed 29 April 2026. All six systems ran to completion (S4: 127/127 segments, 55:39 total). Results below supersede earlier estimates.

Fluency (chrF/BLEU): S2 scores best on both chrF (35.58) and BLEU (19.84), confirming that full-document context improves surface quality. COMET range across all systems is 0.587–0.638 (spread 0.051) — larger than baseline NER run but still modest; neural metric sees limited differentiation between systems.

Terminology (HTM): S2, S3, S4, and S5 Mistral all reach HTM 0.375 under BioMistral NER, up from 0.304 (S1/S5-NLLB). S4 HTM rose from 0.286 (baseline NER, CCR=0.085) to 0.375 (BioMistral NER, CCR=0.354) — directly confirming that CCR is the binding constraint. At CCR=0.354 the reranker finally has signal on a meaningful fraction of segments.

System	BLEU	chrF	COMET	TCR	HTM	CCR	mean s/seg	p95 s/seg
<i>Baseline NER (Jean-Baptiste/camembert-ner, CCR = 0.085)</i>								
S1 NLLB	18.89	31.90	0.638	0.800	0.304	0.085	0.78	4.02
S2 Mistral (doc)	19.71	35.48	0.640	1.000	0.304	0.085	10.74	19.94
S3 GraphRAG	15.34	34.11	0.619	1.000	0.321	0.085	3.84	14.41
S4 rerank	16.29	34.60	0.627	1.000	0.286	0.085	17.89	56.45
S5 NLLB+boost	18.89	31.90	0.638	0.800	0.304	0.085	0.72	3.66
S5 Mistral+boost	16.46	34.85	0.627	1.000	0.321	0.085	4.02	15.43
<i>BioMistral-7B NER (prompting, CCR = 0.354)</i>								
S1 NLLB	18.89	31.90	0.638	0.800	0.304	0.354	0.78	4.02
S2 Mistral (doc)	19.84	35.58	0.624	1.000	0.375	0.354	10.74	19.94
S3 GraphRAG	10.79	32.65	0.589	1.000	0.375	0.354	3.84	14.41
S4 rerank	10.96	32.34	0.592	1.000	0.375	0.354	26.29	69.50
S5 NLLB+boost	17.34	30.55	0.636	0.800	0.304	0.354	0.81	3.66
S5 Mistral+boost	10.82	32.64	0.587	1.000	0.375	0.354	4.95	15.20

Table 4: Evaluation results under two NER conditions. S1/S5-NLLB are unaffected by NER quality (no graph path). COMET = Unbabel/wmt22-comet-da, run on CPU. Inference times are unchanged across NER conditions.

NER model	Type	CCR	Spans extracted	Best downstream HTM
Jean-Baptiste/camembert-ner	General-domain encoder	0.085	58	0.3
CamemBERT EMEA+MEDLINE	Fine-tuned encoder	0.266	433	TB
BioMistral-7B	LLM prompting	0.354	472	0.3

Table 5: NER regime comparison. CCR and downstream HTM both improve with richer NER coverage, confirming that NER quality is the binding constraint on the full pipeline.

TCR: S1 and S5-NLLB score 0.800; all Mistral-based systems reach 1.000. NER condition does not affect TCR because TCR measures gold-term surface presence, not MedDRA-level grounding.

Fluency–terminology trade-off: Under BioMistral NER, the BLEU cost of graph grounding is severe — S3 drops from S2’s 19.84 to 10.79 (−9.05 points) while HTM stays flat (0.375). This confirms the trade-off is real: richer graph context injects correct terminology but disrupts fluency. S5-Mistral+boost matches S3 on both HTM (0.375) and BLEU (10.82) — the logit boost adds no independent benefit over GraphRAG under this NER condition.

Speed: S5-NLLB remains the fastest graph-using system (0.81 s/seg mean), 32× faster than S4 (26.29 s/seg). S4 p95 of 69.5 s/seg reflects 5 sequential Mistral samples per segment at higher CCR. S1 and S5-NLLB are identical on BLEU/HTM/COMET confirming logit boost has no measurable effect via NLLB at this CCR level.

6 Qualitative Error Analysis

6.1 Pneumonitis Cluster (48_040–48_047)

Source: *pneumopathie inflammatoire* / *pneumopathie inflammatoire à médiation immunitaire*.

Reference: pneumonitis (PT, L4) / immune-mediated pneumonitis (LLT, L5).

6.2 Immune-Mediated Modifier Drift

Effets indésirables à médiation immunitaire → reference: “immune-mediated adverse reactions”.

All systems fail on this term. S1 and S5 NLLB produce “immune mediation” (non-standard).

System	Rendering
S1	“inflammatory pneumopathy”, “inflammatory pneumonia”
S2	“interstitial lung disease” (different MedDRA concept, HLT L3)
S3/S4/S5 Mistral	“immune-mediated pneumopathy” (non-canonical)
S5 NLLB	“inflammatory pneumopathy”

Table 6: No system produces the canonical MedDRA preferred term without hard constraints.

S2, S3, S4, and S5 Mistral produce “immunomodulatory” (semantically broader and MedDRA-incorrect). No system produces the canonical form across any of the 127 segments. This is the most frequent gold term and the most consistent failure.

6.3 S3-Specific Failures: Deep Qualitative Review

A full segment-level review of the NER-refreshed S3 run identified the following failure categories.

6.3.1 GraphRAG Context Leakage into Hypothesis

Approximately six segments show the model treating retrieved MedDRA context as part of its own output rather than as a reference. Examples:

- 48_039: hypothesis ends with “(MedDRA LLT L5: SLE — Recommendations)”
- 48_044: inline meta-comment “Inflammatory pneumopathy is not a specific term in MedDRA...”
- 48_056, 48_060, 48_089: appended notes of the form “MedDRA term: Hepatitis, resolved / Note: The term...”
- 48_120: hypothesis *begins* with “SLE for: ‘Les’ (MedDRA LLT L5)” before the translation

This pattern is absent in S1 and S2 and is specific to GraphRAG. It represents a prompt engineering failure: the model is not keeping retrieved context and generation target separate. In a production pipeline all such annotations would corrupt downstream processing.

6.3.2 New Repetition Loop

48_024 (frequency definition segment): “Very frequent ($\geq 1/10$): SLE.1.1.1.1.1...” repeated hundreds of times. S1 loops occurred on long tabular segments; S3’s loop occurred on a short definitional segment. The trigger condition is different, confirming that repetition instability is not purely length-dependent.

6.3.3 Confabulated Training Examples

48_003: after the correct translation, the model appended four fabricated continuation examples (“Les données de l’étude sont issues de...”) resembling few-shot training data. The GraphRAG context appears to have activated a pattern-completion mode. Not a loop but a distinct confabulation failure.

6.3.4 Acronym Hallucination

Low-frequency acronyms not covered by NER spans are confabulated:

LHc is particularly notable: it renders as “Lymphopenia with Hypoalbuminemia” in some segments and “body weight less than or equal to the healthy child” in others. The inconsistency confirms the model has no stable mapping and is generating plausible-sounding expansions from partial character matches.

Source	S3 output	Correct
CBNPC	Central Nervous System Cranial Nerve Palsies	NSCLC
GCSH	CAR-T therapy	Allogeneic HSCT
LHc	Lymphopenia with Hypoalbuminemia / “healthy child”	cHL
CETEC	Colorectal cancer	HNSCC
SSJ/NET	Hypersensitivity Shock Syndrome and Neutropenia	SJS/TEN

Table 7: Acronym hallucinations in S3. LHc produces different wrong expansions across segments, confirming non-grounded confabulation.

6.3.5 Consistent Surface Errors Across All Systems

These errors appear in S3 but are not S3-specific:

- *platine* → “platine-based” (should be “platinum-based”) — all systems
- *RCP* → “product information” (should be “SmPC”) — all systems
- *ITK* not corrected to “TKI” — S1, S3, S4, S5
- *RCT/radiochimiothérapie* → “RT-CT” (should be “CRT”) — S3
- *syndrome main-pied* left untranslated (should be “palmar-plantar erythrodysesthesia syndrome”) — S1, S3, S4, S5

6.3.6 S3 Failure Summary

Category	Segments	Examples
GraphRAG context leakage	≈6	48_039, 48_044, 48_056, 48_060, 48_089, 48_120
Repetition loop	1	48_024
Confabulated training examples	1	48_003
Acronym hallucination	≈8	CBNPC, GCSH, LHc, CETEC, SSJ/NET
Immune-mediated modifier wrong	All applicable	48_001, 48_009, ...
Pneumonitis non-canonical	8	48_040–48_047

Table 8: S3 failure categories from full segment review (NER-refreshed run).

7 Engineering Failures and Fixes

7.1 Repository Structure (post-cleanup, 30 April 2026)

The repository was restructured for clarity after all experiments completed. The canonical layout is:

```
MT_Project_Terminology/
  pipeline/           core pipeline (was src/ + metrics/)
  systems/           S1-S5 implementations
  graph.py           Neo4j MedDRA grounding
  planning.py        Terminology planning locks
  metrics/           HTM, TCR, CCR
  scripts/           main entry points
  run_pipeline.py     (was run_systems.py)
  evaluate.py
  plot_results.py
  prepare_data.py
  experiments/       supplementary only -- not main paper
  ner_ablation/      NER model comparison scripts + results
```


vector_grounding/	Vector CCR experiments + results
results/	PAPER RESULTS -- do not move
ner_baseline/	Jean-Baptiste NER (CCR 0.085)
ner_finetuned/	CamemBERT fine-tuned (CCR 0.266)
ner_biollm/	BioMistral zero-shot (CCR 0.354)
data/section48/	segments, gold terms, NER outputs
models/	trained checkpoints
rerun_all.sh	reproduce everything
README.md	what / reproduce / results

All 7 unit tests (`tests/test_metrics.py`) pass after restructure. The `libnvjitlink.so.13` library fix required adding `.venv/lib/python3.14/site-packages/nvidia/cu13/lib` to `LD_LIBRARY_PATH` — add to `~/.bashrc` permanently.

Two runtime warnings observed during the BioMistral NER pipeline run (S4, April 2026) that are not yet resolved and warrant further investigation.

Neo4j ambiguous concept matches. During the S4 run, `neo4j._sync.work.result` emitted repeated `UserWarning: Expected a result with a single record, but found multiple`. This indicates that certain French NER terms match more than one MedDRA concept node in the graph — i.e. the `MATCH (c:Concept {fr_label: $fr_term})` Cypher query returns multiple records. The current implementation silently takes the first match, which means the concept selected may not be the most specific or contextually appropriate one. This could depress HTM scores by grounding terms at the wrong MedDRA level.

Investigation needed: export the full list of terms triggering this warning by adding a `DEBUG` log of `fr_term` whenever `result.single(strict=False)` returns more than one record. Likely candidates are short, high-frequency terms (e.g. *pneumonie*, *infection*) that appear at multiple levels of the MedDRA hierarchy. A disambiguation step — selecting the concept whose `level` most closely matches the source segment’s expected specificity — would improve grounding quality.

Table 2 segmentation (segment 48_028). The MedDRA adverse reactions table (Tableau 2, ~5–6 pages of the 10-page Section 4.8) is treated as a single segment (48_028) by the sentence segmenter. This is the largest and most terminology-dense segment in the corpus — it contains hundreds of MedDRA-listed adverse events across three treatment columns and five frequency categories. The segment caused parse failures or truncations in every NER model tested. Because all systems are evaluated against the same 127 segments including this segment, cross-system comparisons remain valid. However, the overall CCR and HTM scores are conservatively underestimated: splitting the table into individual SOC/frequency rows would produce hundreds of short, high-precision segments where NER extraction and MedDRA grounding would be far more reliable. This is noted as a corpus limitation rather than a pipeline failure. Proper table-aware segmentation is left as future work. Switching from baseline NER segments to BioMistral NER segments triggers a `WARNING: planning_locks.json may be stale` on every run, because the locks were computed on a different NER span set. For the current ablation this is acceptable — each NER condition ideally uses freshly computed locks — but the staleness check only warns rather than blocking. For final paper results, each of the three NER conditions should have its own dedicated planning locks file (e.g. `planning_locks_baseline.json`, `planning_locks_finetuned.json`, `planning_locks_biollm.json`) so results are fully reproducible without manual deletion.

8 NER Ablation Design

8.1 Motivation

The NER stage is the binding constraint on the full pipeline. CCR rising from 0.085 to 0.354 (BioMistral zero-shot) produced a +0.054 absolute HTM improvement across all Mistral-based systems. This motivates a systematic ablation of NER approaches independently of the MT systems.

8.2 Relevant Prior Work

In a separate internship project (HiTZ, UPV/EHU) evaluating LLMs for clinical text neutralisation using the CasiMedicos dataset, instruction-tuned Mistral-7B outperformed MedLLaMA on structured medical tasks despite MedLLaMA’s domain-specific pre-training. The task there was medical argumentation and neutralisation — distinct from NER — but the finding motivates testing instruction-following quality as an independent variable from biomedical pre-training.

8.3 Mistral-7B-Instruct Result (Confirmed)

Mistral-7B-Instruct-v0.2 zero-shot NER achieved CCR = **0.222** (225/1012 spans grounded), compared to BioMistral-7B’s 0.354. This *reverses* the internship finding and confirms the hypothesis stated above: biomedical pre-training matters more for NER span extraction than instruction-following quality alone.

Two failure modes were observed. First, four segments (48_023, 48_029, 48_032, 48_125) produced unparseable JSON because Mistral-Instruct hallucinated very large term lists (50+ items) that exceeded the output buffer and truncated mid-array. BioMistral, with stronger domain priors, extracts more targeted term sets. Second, many extracted terms were anatomical or procedural rather than MedDRA adverse event concepts, reducing grounding rate. This suggests the instruction-following advantage of Mistral-Instruct does not compensate for its lack of French biomedical terminology knowledge on this specific extraction task.

8.4 Ambiguous MedDRA Terms (Confirmed)

The `compare_grounding.py` run identified 10 terms in string mode that map to multiple MedDRA concepts. The most frequent ambiguous terms are: *fatigue* (5 occurrences, 2 concepts), *pneumopathie inflammatoire* (8 occurrences, 2 concepts), *toux* (3 occurrences, 2 concepts), *vomissement* (3 occurrences, 2 concepts), *hypotension orthostatique* (2 occurrences, 3 concepts), *syncope* (1 occurrence, 3 concepts), *agranulocytose* (1 occurrence, 3 concepts), *perte de poids* (1 occurrence, 3 concepts).

These are structurally ambiguous entries in MedDRA itself — short, high-frequency French terms that appear at multiple levels of the hierarchy (e.g. *fatigue* exists as both an LLT and a PT). Vector grounding (once embeddings are built) should resolve these by selecting the concept whose semantic neighbourhood best matches the source sentence context.

8.5 Hypothesis

Two capabilities determine LLM NER quality on this task:

1. **Biomedical knowledge:** recognising French medical terms as entities (domain-specific pre-training helps here)
2. **Instruction following:** reliably returning a well-formed JSON list (instruction tuning helps here)

These are orthogonal. A well instruction-tuned general model may outperform a biomedical model with weaker instruction following, particularly on output format compliance. NER span extraction is more precision-sensitive than generation tasks, so biomedical pre-training may matter more here than in the internship setting.

8.6 NER Comparison Results — Complete (29 April 2026)

8.7 Vector Grounding Results (30 April 2026) — Complete

8.8 Key Interpretation: String CCR Measures Surface Form, Not Conceptual Quality

The vector CCR results (30 April 2026) fundamentally reframe the NER ablation findings. At threshold $\tau = 0.75$, every NER model achieves vector CCR between 0.862 and 0.999 — essentially all extracted terms have a semantically close MedDRA concept in the graph. The large string CCR differences (0.052–0.354) are entirely surface form failures, not conceptual failures.

This means string CCR has been measuring whether the NER model happens to produce the same surface form as MedDRA’s French labels, not whether it extracted a conceptually valid medical term. The practical consequence: the baseline Jean-Baptiste encoder (string CCR 0.052) actually extracts terms that are conceptually grounded 86% of the time — its low string CCR reflects the gap between general NER vocabulary and MedDRA’s specific French label strings.

What this means for the pipeline. Under string grounding, only 35.4% of BioMistral spans feed into the planning and constrained decoding steps. Under vector grounding at $\tau = 0.75$, 98.7% would. This means the HTM improvements from S2–S5 observed under string grounding are achieved with only a third of the available terminology signal. Vector grounding would likely produce substantially higher HTM scores across all systems by giving the planner a much richer set of locked renderings.

Ranking stability. Under vector grounding the models are essentially indistinguishable on CCR (0.862–0.999). The ranking differences visible in string CCR reflect surface form canonicalisation quality, not NER capability. The key differentiator between models under vector grounding is **span count**: Mistral-Instruct extracts 1,012 spans (highest recall) vs Jean-Baptiste’s 58 (lowest). Under vector grounding higher recall means more graph signal, making Mistral-Instruct potentially the strongest model for this pipeline — the opposite of the string CCR ranking.

Connecting to the Technalia thesis. This finding directly motivates the ontology-grounded consistency metric being developed for the Technalia XAI toolbox. String matching against ontology labels systematically underestimates extraction quality for models that use synonyms, paraphrases, or domain-adjacent vocabulary. A robust evaluation framework for LLM clinical explanations must use semantic grounding rather than surface string matching. The vector CCR result here is a concrete empirical demonstration of that gap on real MedDRA data.

BioMistral zero-shot remains the best single model. CCR 0.354 with 472 spans — compact, high-precision extraction. Llama3 extracts more spans (485) but grounds fewer proportionally (CCR 0.326), suggesting higher recall but lower precision — it includes terms that don’t appear in MedDRA.

Instruction following vs biomedical knowledge. The Mistral-Instruct result (CCR 0.222, 1012 spans) is the clearest finding: high instruction-following quality without biomedical pre-training produces high-volume but low-precision extraction — more than double the spans of BioMistral but worse CCR. The model extracts anything that looks like a term rather than

specifically medical concepts. This does not replicate the HiTZ internship finding; the task structure difference (span extraction vs generation) explains the divergence.

LoRA smoke run — degenerate output. BioMistral LoRA trained on 64 records produced CCR 0.075 with 2678 spans — the model entered repetition loops (e.g. “traitement médical” repeated 20+ times, “cancer” repeated 40+ times) and on one segment produced a French paragraph praising its own performance instead of a term list. This is classic underfitting: 64 training records is insufficient to teach the EOS boundary. The 2678 spans vs BioMistral’s 472 confirms the model is generating unconstrained repetitions rather than extracting entities. Full QUAERO training (all EMEA + MEDLINE sentences, ~3,000+ annotated sentences) is required for valid results. Demoted to **future work**.

Baseline encoder anomaly. The Jean-Baptiste baseline CCR of 0.052 (3/58 spans, 127 segments) is lower than the previously reported 0.085. The span count of 58 across 127 segments (≈ 0.46 spans/segment) suggests `segments_ner_baseline.jsonl` may have been regenerated with a higher confidence threshold or fewer entity classes enabled. The 0.085 figure from earlier runs on the same model is the more reliable baseline and should be used in the paper.

Summary ordering. By CCR: BioMistral zero-shot (0.354) > Llama3 (0.326) > CamemBERT fine-tuned (0.266) > Mistral-Instruct (0.222) > Jean-Baptiste (0.085) $\gg$ LoRA smoke (0.075). The ranking confirms that biomedical pre-training with moderate instruction following (BioMistral) is the optimal operating point for this task given current resources.

Llama3:8b zero-shot via Ollama achieves CCR 0.392 (235/599 spans grounded), exceeding BioMistral-7B zero-shot (CCR 0.354, 472 spans). This is a partial replication of the HiTZ internship finding: a strong general instruction-following model can match or exceed a biomedical-specific model even on clinical extraction tasks, provided instruction quality is high enough. Llama3’s RLHF training produces more reliable JSON output than BioMistral’s weaker instruction following (0 parse retries vs BioMistral’s occasional format failures).

Note the denominator difference: Llama3 produced 599 NER spans vs BioMistral’s 472. Llama3 extracts more candidate terms per segment, increasing both the numerator (more terms to ground) and denominator (more total spans). The CCR ratio is higher but the absolute grounded count (235 vs 167) also improves. This suggests Llama3’s instruction following produces higher recall extraction.

The Mistral-7B-Instruct result (CCR 0.222) remains the clearest negative: high instruction-following quality without biomedical knowledge is insufficient. Llama3’s advantage over Mistral-Instruct likely comes from its larger parameter count (8B vs 7B) and stronger RLHF alignment rather than biomedical knowledge specifically.

8.9 Vector Grounding: Still Invalid Despite Index Being Built

The 29 April overnight run successfully built the Neo4j vector index (93,185 concepts embedded using `paraphrase-multilingual-mpnet-base-v2` in 3:12). However `compare_grounding.py` still reports vector CCR = 1.000. The bug is in the similarity threshold: the vector index query returns a result for every NER span regardless of similarity score, because the fallback behaviour on low-confidence matches returns the nearest neighbour unconditionally rather than rejecting below threshold.

Fix needed in `compare_grounding.py`: after the vector query, check `score >= 0.6` (tunable) before accepting the grounded concept. If `score < 0.6`, fall back to string match or mark as ungrounded. The current 0-fallback count confirms no spans are being rejected.

Additionally S5_mistral under vector grounding reported HTM 0.286 (down from 0.375 under string grounding) — this is a direct consequence of the CCR=1.000 bug: when every span is "grounded" including incorrect ones, HTM scoring receives wrong level assignments and drops. Vector grounding results remain invalid until the threshold fix is applied.

8.10 Outstanding Issues from 29 April Run

- **Unsloth:** requires `pip install unsloth_zoo` in addition to the main package. Run `pip install unsloth_zoo -break-system-packages` then retry `finetune_ner_unsloth.py`.
- **S3 vector OOM:** Llama3 was still resident in GPU memory when S3 tried to load Mistral-7B for the vector grounding run. Fix: add `torch.cuda.empty_cache()` and explicit model deletion between NER and MT pipeline steps in the chain script.
- **Vector CCR threshold:** add `-vector-threshold` argument to `compare_grounding.py` (default 0.6); reject spans below threshold and report fallback count.

The confirmed Mistral-7B-Instruct CCR of 0.222 vs BioMistral CCR of 0.354 is a clean negative result relative to the hypothesis drawn from prior internship work (HiTZ, UPV/EHU), where instruction-tuned general models outperformed domain-specific models on medical argumentation and neutralisation tasks. That finding does *not* replicate for French clinical NER.

The difference is task structure. Argumentation and neutralisation are generation tasks where instruction-following quality (output format, register, length) dominates. NER is a span extraction task requiring precise knowledge of *what counts as a medical entity in French regulatory text* — biomedical pre-training provides that knowledge; instruction tuning does not. This distinction is a contribution in itself.

Mistral-Instruct additionally produced unparseable JSON on 4/127 segments (48_023, 48_029, 48_032, 48_125) where the model output exceeded the token limit and truncated mid-list. BioMistral produced zero truncations under the same settings. This suggests BioMistral’s medical pre-training also improves output concision on clinical terminology lists.

8.11 Evaluation Integrity Gates

A key methodological improvement from the overnight run is the identification of silent failure modes that produced misleading results:

- **Missing vector index** → fake CCR = 1.000. Fixed with preflight check.
- **Partial S4 run** (3/127 segments evaluated) → nonsense BLEU 18.66 / chrF 58.53 / HTM 0.000. Fixed with row-count preflight in `evaluate.py`.
- **NER token truncation** → 4 segments silently dropped. Fixed with retry at temperature=0.
- **Ollama 404** → empty JSONL written silently, CCR reported as 0.000. Fixed with explicit error exit.

The broader principle is that apparent gains from more complex systems frequently dissolve under strict evaluation. The evaluation integrity layer is itself a contribution to reproducibility.

8.12 Revised Ablation Hierarchy

The overnight results confirm a three-tier ordering of what drives pipeline performance:

Tier 1 (dominant) NER quality and concept grounding correctness. CCR is the key diagnostic. All system-level differences are second-order to this.

Tier 2 (important) Retrieval method (string vs vector). Vector grounding remains unvalidated but is the correct architectural direction.

Tier 3 (marginal once Tier 1 fixed) Reranking vs logit constraints vs prompt structure. S3/S4/S5 differences are small once grounding is adequate.

This hierarchy is the central claim of the paper: *increasing MT system complexity yields diminishing returns unless upstream terminology extraction is reliable.*

The LoRA fine-tuning uses Unsloth for memory-efficient QLoRA (rank 16, $\alpha = 32$) on BioMistral-7B. Training data is QUAERO EMEA converted to Alpaca instruction format: input = French sentence, output = JSON term list. Loss is computed on responses only (`train_on_responses_only`), which is critical for output format quality.

9 Final Results — Three NER Conditions (Confirmed 30 April 2026)

9.1 Complete Three-Condition Results Table

9.2 Key Findings from the Three-Condition Table

NER is the binding constraint — confirmed. The clearest result is the HTM progression for S2 (full-document Mistral, no graph):

- Baseline NER (CCR 0.085): HTM 0.304
- Fine-tuned NER (CCR 0.266): HTM 0.375 (+0.071)
- BioMistral NER (CCR 0.354): HTM 0.375

HTM jumps from 0.304 to 0.375 between baseline and fine-tuned NER, then plateaus. The improvement is entirely attributable to NER quality, not to S2’s architecture — S2 has no graph component.

Fine-tuned NER gives the best BLEU overall. S2 under fine-tuned CamemBERT achieves BLEU 21.02 — the highest BLEU across all 15 system/NER combinations. Better NER provides the document-context system with cleaner terminology anchors, improving both fluency and terminology simultaneously.

Fluency–terminology trade-off sharpens at high CCR. Under BioMistral NER, S3/S4 BLEU drops to 10.79/10.96 while HTM holds at 0.375. The graph context injected at CCR=0.354 is dense enough to improve terminology but disrupts fluency. Under fine-tuned NER (CCR=0.266), S3/S4 BLEU is 15.95/16.37 — the trade-off is present but less severe.

S4 is no longer a negative result at adequate CCR. Under baseline NER (CCR=0.085) S4 HTM was 0.286 — the worst result. Under fine-tuned NER S4 HTM rises to 0.339, matching S3. At BioMistral CCR=0.354 S4 HTM reaches 0.375. The reranker requires sufficient graph signal to function — below CCR $\approx$ 0.20 it is worse than no reranking; above it converges to S3 performance.

S1 and S5-NLLB are NER-invariant. As expected: NLLB-200 with no Mistral path is completely unaffected by NER condition. BLEU 18.89, HTM 0.304 in all three rows. This is the clearest confirmation that any HTM variation across rows is causal, not statistical noise.

9.3 Engineering Complete — Paper Remaining

All experiments are done. The engineering log is complete at 16 pages. The remaining work is:

1. Run S5 Mistral+boost on fine-tuned NER segments to fill the one missing cell

2. Manual error analysis with partner (20 segments, Cohen's kappa)
3. Write the paper (26 days remaining)

Do not run any further experiments. The story is complete.

Issue	Root cause	Fix
S4 OOM (3 crashes)	Batched KV cache with 5 sequences; zombie process holding VRAM; caching allocator warmup	Sequential generate(); kill zombie; <code>expandable_segments=True</code>
Model reload per segment	<code>load_mistral_4bit()</code> inside segment loop	Hoisted to <code>run()</code>
Partial output overwritten	No resume logic	<code>-resume</code> flag; append mode JSONL
Neo4j build 40+ min	MERGE without index = $O(n^2)$	RANGE indexes on <code>id</code> and <code>fr_label</code> ; build now <2 min
CCR = 0 (stage 1)	neo4j driver not installed	<code>pip install neo4j</code> ; rebuild graph
CCR = 0 (stage 2)	<code>almanach/camembert-bio-base</code> has no NER head; all scores ≤ 0.58	Switched to <code>Jean-Baptiste/camembert-ner</code> ; 58 spans extracted
TCR vacuously 1.000	“no EN hit” counted as <code>None</code> , excluded from denominator	Gold terms with triggers but zero EN hits now score 0.0
COMET install fail	<code>unbabel-comet</code> pins <code>numpy<2</code> ; no wheel for Py 3.14, build fails on missing <code>Python.h</code>	Installed <code>unbabel-comet==2.2.7</code> with <code>-no-deps</code> ; explicit deps keep NumPy 2.x; compat shim for <code>functools._HashedSeq</code> in <code>lru_cache.py</code>
COMET on Py 3.14	<code>comet</code> imports <code>functools._HashedSeq</code> removed in Py 3.13+	One-line shim at top of <code>comet/models/lru_cache.py</code> ; model <code>Unbabel/wmt22-comet-da</code> loaded successfully
Inference plot negative x-axis	<code>xerr</code> was symmetric, extending left by $p95 - \mu$; axis dipped below 0	Asymmetric <code>xerr = (0, p95 - mean)</code> ; <code>ax.set_xlim(left=0)</code>
S4 reranker no-op	<code>score_candidate()</code> returned 0.0 on zero grounded spans, so all candidates scored equally and <code>cands[0]</code> always won silently	Returns <code>None</code> on zero spans; <code>translate_segment()</code> skips rerank and logs DEBUG message
S3/S4 no token truncation	<code>build_mistral_prompt()</code> had no length guard; long segments (e.g. 48_028: 2523 tokens) silently exceeded NLLB 1024-token limit	Import <code>truncate_full_section_for_token_budget</code> ; apply before prompt build in both S3 and S4
Stale planning cache	<code>planning_locks.json</code> reused silently regardless of whether <code>segments_ner.jsonl</code> changed	Store <code>_segments_mtime</code> on write; warn on mtime mismatch > 1s on read
<code>plot_results.py</code> crash	<code>TermGraph._ambiguous_lines</code> attribute referenced before assignment in <code>reset_grounding_stats()</code>	Rename to <code>_ambiguous_norms</code> or initialise in <code>__init__</code> ; figures generated by running <code>evaluate.py</code> directly
Vector CCR = 1.000 (fake)	<code>compare_grounding.py</code> ran vector path without checking whether Neo4j vector index existed; silently reported perfect CCR	Preflight: <code>SHOW INDEXES YIELD name</code> checks for <code>meddra_fr_embedding</code> ; fallback to <code>CALL db.indexes()</code> ; exit 1 if missing
Partial S4 evaluation	<code>evaluate.py</code> scored S4 on 3/127 rows,	Row-count pre-

Model	Biomed	Instruction	Training	CCR	Spans
Jean-Baptiste/camembert-ner	Low	N/A	None	0.052	58
CamemBERT EMEA+MEDLINE	Low	N/A	QUAERO BRAT	0.266	—
Mistral-7B-Instruct zero-shot	Low	High	None	0.222	1012
BioMistral-7B zero-shot	High	Medium	None	0.354	472
Llama3:8b zero-shot (Ollama)	Medium	V. high	None	0.326	485
BioMistral-7B LoRA (64 rec.)	High	High	QUAERO (smoke)	0.075	2678

Table 10: Complete NER ablation. All CCR values confirmed on `segments_ner_*.jsonl` against MedDRA string grounding. “Spans” = total NER spans extracted across 127 segments. LoRA trained on 64 records only (smoke run) — see below.

Model	String CCR	Spans	Vec CCR ($\tau=0.75$)	Grounded	Fallbacks
Jean-Baptiste baseline	0.052	58	0.862	50	8
CamemBERT fine-tuned	0.266	433	0.998	432	1
Mistral-7B-Instruct	0.222	1012	0.999	1011	1
Llama3:8b zero-shot	0.326	485	0.992	481	4
BioMistral LoRA full	0.324	580	0.998	579	1
BioMistral-7B zero-shot	0.354	472	0.987	466	6

Table 11: String vs vector CCR at cosine threshold $\tau = 0.75$. All values computed on Section 4.8 segments (127 segments, string grounding = exact `fr_label` match; vector grounding = nearest-neighbour cosine similarity in Neo4j against 93,185 embedded MedDRA concept nodes). All CCR numbers are on BioMistral NER segments except where the model has its own segment file.

NER	System	BLEU	chrF	COMET	TCR	HTM	CCR
Baseline CCR=0.085	S1 NLLB	18.89	31.90	0.638	0.800	0.304	0.085
	S2 Mistral doc	19.71	35.48	0.640	1.000	0.304	0.085
	S3 GraphRAG	15.34	34.11	0.619	1.000	0.321	0.085
	S4 rerank	16.29	34.60	0.627	1.000	0.286	0.085
	S5 NLLB+boost	18.89	31.90	0.638	0.800	0.304	0.085
	S5 Mistral+boost	16.46	34.85	0.627	1.000	0.321	0.085
Fine-tuned CamemBERT CCR=0.266	S1 NLLB	18.89	31.90	0.638	0.800	0.304	0.266
	S2 Mistral doc	21.02	36.63	0.641	1.000	0.375	0.266
	S3 GraphRAG	15.95	35.22	0.628	1.000	0.339	0.266
	S4 rerank	16.37	35.48	0.630	1.000	0.339	0.266
	S5 NLLB+boost	17.27	30.48	0.636	0.800	0.304	0.266
	S5 Mistral+boost			pending			
BioMistral zero-shot CCR=0.354	S1 NLLB	18.89	31.90	0.638	0.800	0.304	0.354
	S2 Mistral doc	19.84	35.58	0.624	1.000	0.375	0.354
	S3 GraphRAG	10.79	32.65	0.589	1.000	0.375	0.354
	S4 rerank	10.96	32.34	0.592	1.000	0.375	0.354
	S5 NLLB+boost	17.34	30.55	0.636	0.800	0.304	0.354
	S5 Mistral+boost	10.82	32.64	0.587	1.000	0.375	0.354

Table 12: Complete three-condition results. All values on 127 segments of Keytruda Section 4.8. S1 and S5-NLLB are identical across NER conditions (no Mistral path). S5 Mistral fine-tuned pending one additional run.